

EN-GB

Adults



CL!MATE FRESK

All the cards are in your hands!

How to play

You need one deck of cards per team (4 to 7 people), a 1 x 2 m piece of paper, pencils, rubbers, coloured pens and some tape.

The aim of the game is for each team to map out the story of climate change by arranging the cards on the table, in order of cause and effect, and drawing arrows to show links between cards.

Wait for the team to position all the cards they have before giving them the next set.

Timings: 90 mins to play the card game; 20-30 mins to decorate and present the Fresk; 10 mins to share feelings; and 45-60 mins to discuss action ideas.



Understanding



**Creativity &
Summary**



Emotions



Discussion

For a simpler version of the game, take out cards 10, 14, 15 and 41.

Human activities

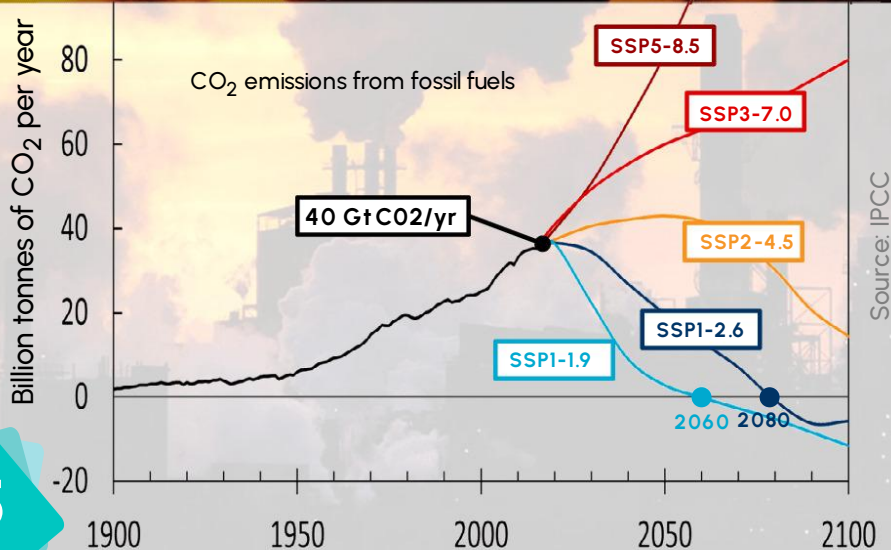
1



This is where it all begins...

Set 1

Burning Fossil Fuels

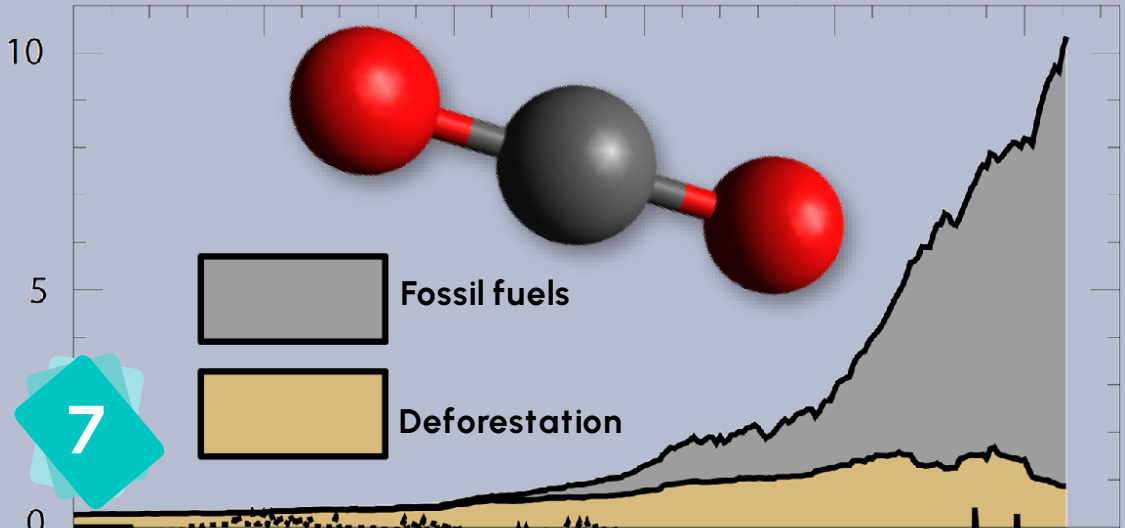




Fossil fuels are coal, oil, and natural gas. They are used mainly in buildings, transportation, and industry. They emit CO₂ when burned.

CO₂ Emissions

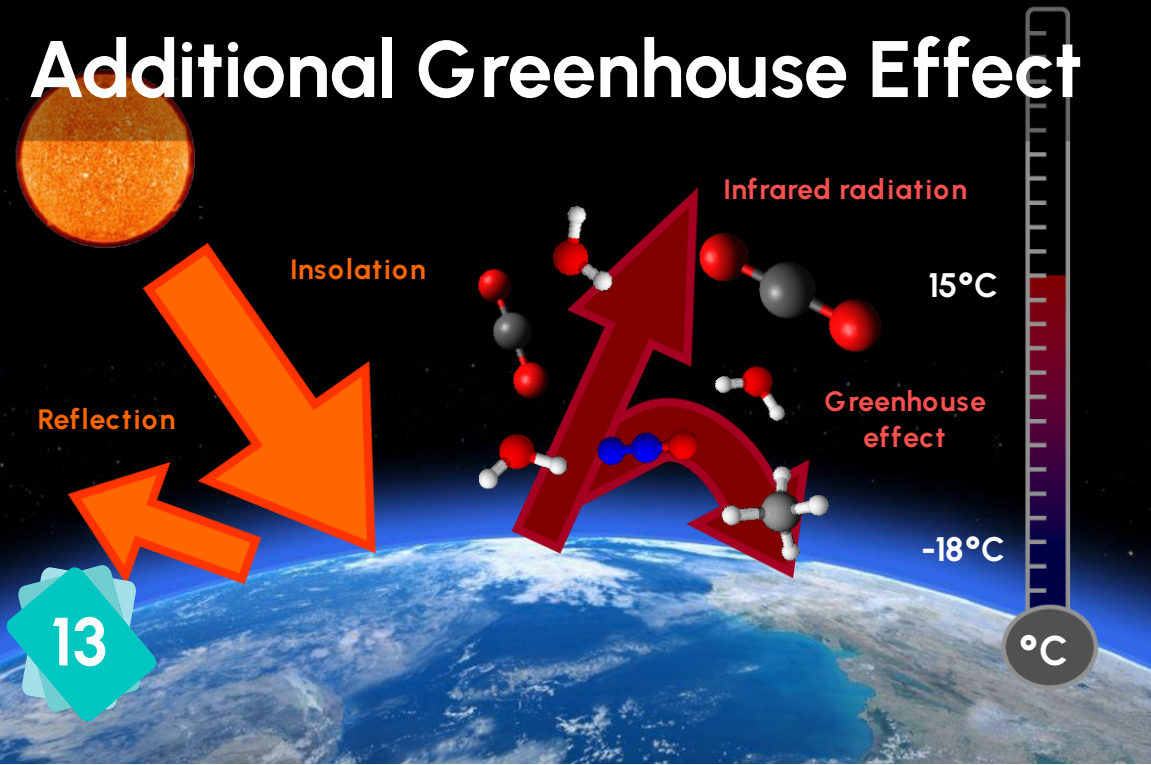
Anthropogenic CO₂ emissions (GtC/yr)





Carbon dioxide (CO_2) is the main anthropogenic greenhouse gas (i.e. caused by human activities). CO_2 emissions come from burning fossil fuels and deforestation.

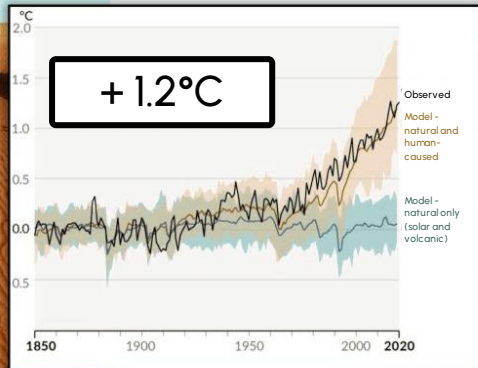
Additional Greenhouse Effect



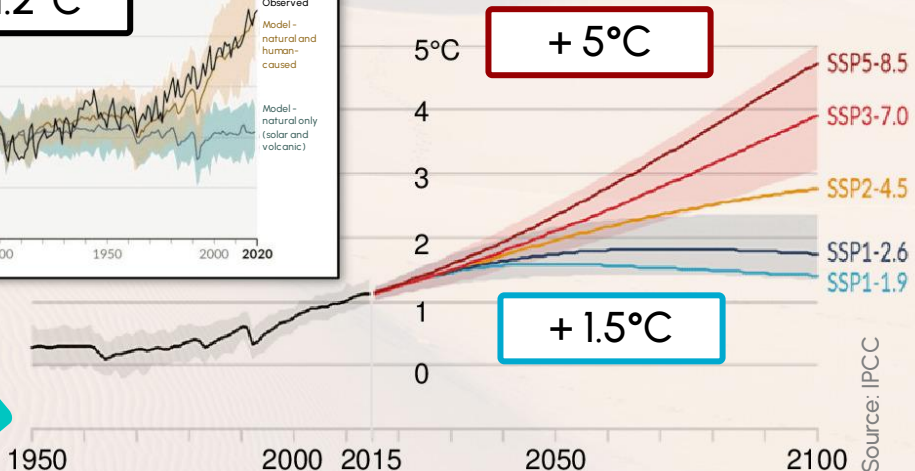


The greenhouse effect is a natural phenomenon. In fact, water vapour is the most common GHG. Without the greenhouse effect, the planet would be 33°C colder. CO₂ and other GHGs from human activities amplify the greenhouse effect which warms up the climate.

Rising Temperatures



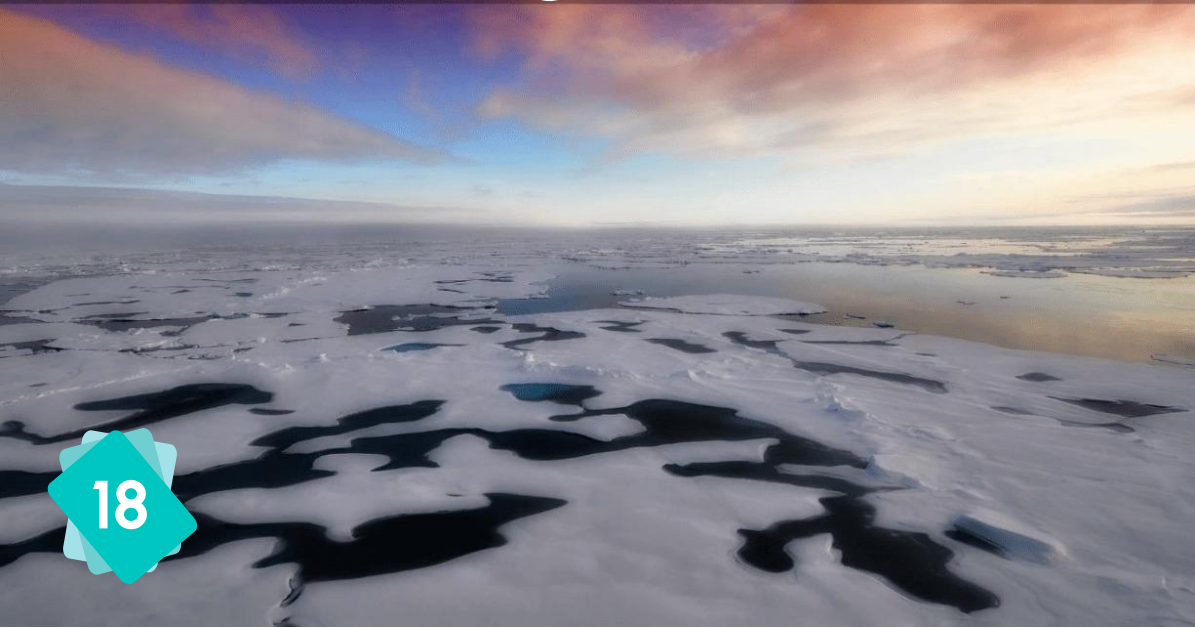
Global surface temperature change
from 1850-1900





The average air temperature at the surface of the Earth has increased by 1.2°C since 1900. Depending on our future emissions, temperatures are predicted to rise by between 1.5°C and 5°C by 2100. At the peak of the last ice age 20,000 years ago, the average temperature was only 5°C lower than today, and it took 10,000 years for the Earth to warm up!

Melting Sea Ice



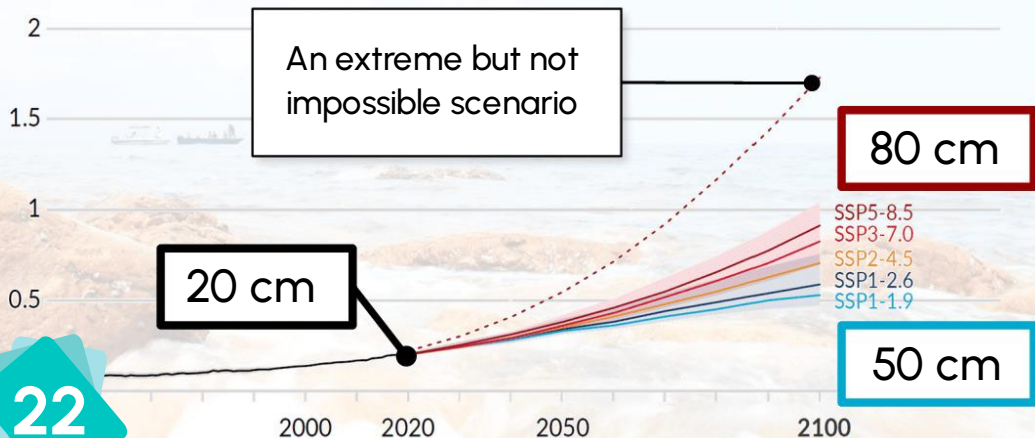
18



Melting sea ice does not make the sea level rise (just like a melting ice cube that is floating in a drink does not make the glass overflow).

Sea Level Rise

Average global sea level rise since 1900



Source: IPCC



Since 1900, sea levels have risen by 20cm. This is caused by ocean water expanding as it heats up, and glaciers and ice sheets melting.

Industry





Industry uses fossil fuels and electricity. It accounts for 40% of greenhouse gas (GHG) emissions.

Building Usage





Houses and commercial buildings
use fossil fuels and electricity.
This accounts for 20% of greenhouse
gas (GHG) emissions.

Transportation





The transportation sector is highly dependent on oil. It accounts for 15% of greenhouse gas emissions.

Deforestation





Deforestation is defined as cutting down or burning more trees than the forest can replace. 80% of deforestation is driven by agricultural expansion. As well as emitting CO₂, it destroys the natural habitats of many species.

Agriculture

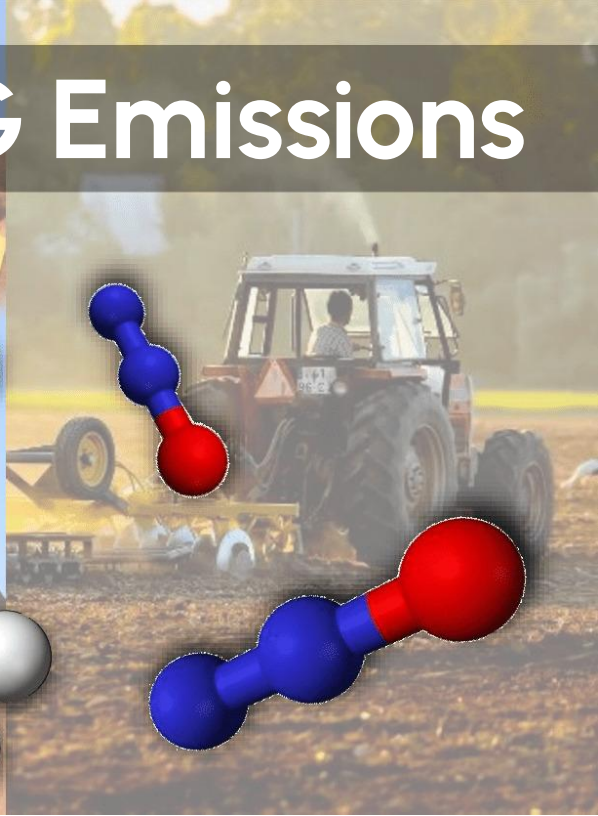
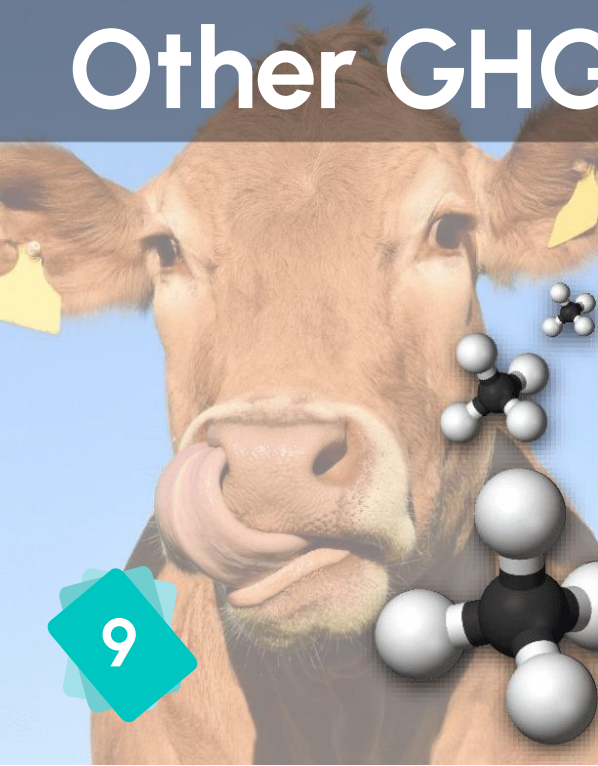




Agriculture is responsible for emitting CO₂ (through deforestation), methane (from cattle and rice paddies) and nitrous oxide (from fertilisers). Altogether, agriculture accounts for 25% of greenhouse gases.

Other GHG Emissions

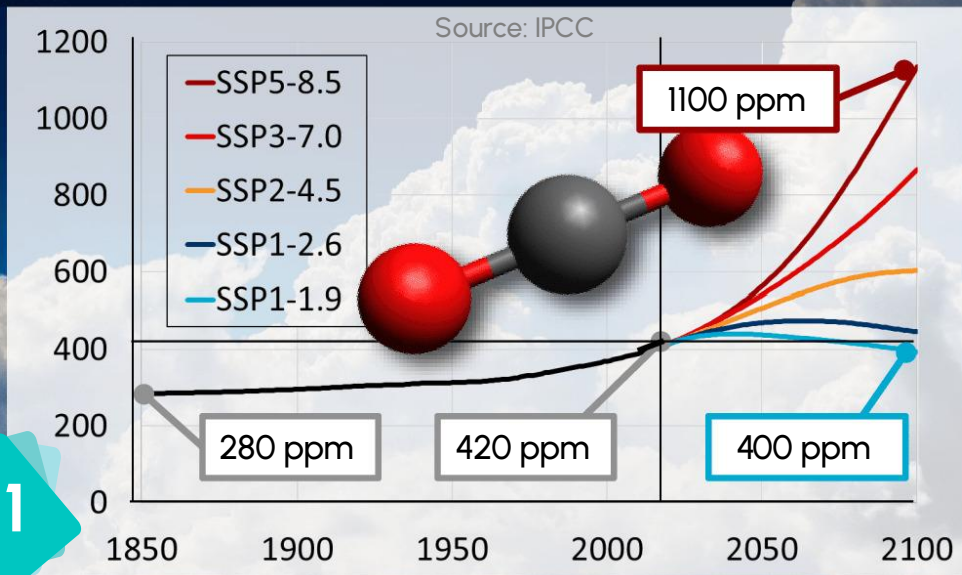
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CO₂ is not the only greenhouse gas (GHG). Among others are methane (CH₄) and nitrous oxide (N₂O), both of which mainly come from agricultural activities.

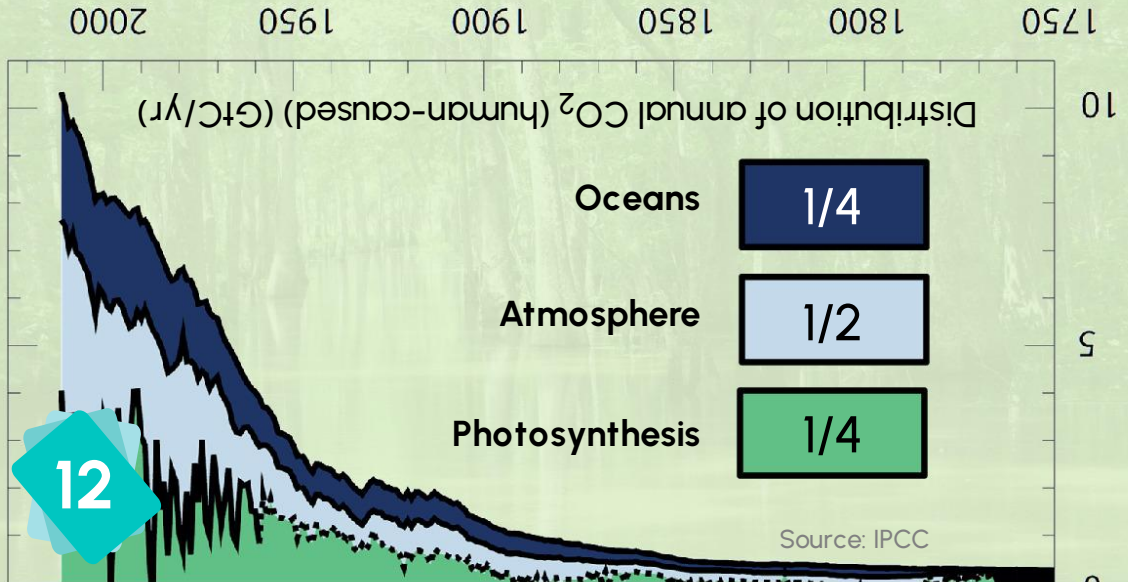
Concentration of CO₂ (ppm)





About half of our CO₂ emissions are captured by natural carbon sinks. The other half stays in the atmosphere. The concentration of CO₂ in the atmosphere has increased from 280 to 420 ppm (parts per million) over the past 150 years.

Carbon Sinks





Half of the CO₂ we emit every year is absorbed by carbon sinks:

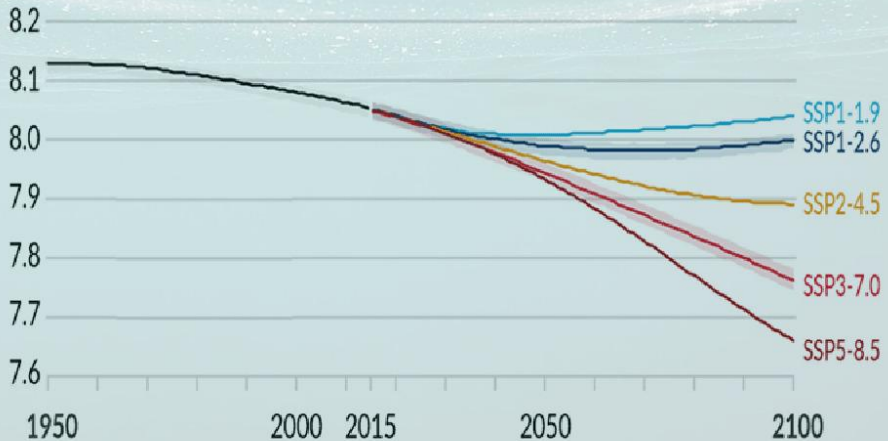
- 1/4 into vegetation via photosynthesis
- 1/4 into the oceans

The remaining half stays in the atmosphere.

Ocean Acidification

Ocean acidification according
to RCP scenarios (pH)

Source: IPCC





When CO_2 dissolves into the ocean,
it turns into ions that acidify the
ocean (the pH drops).

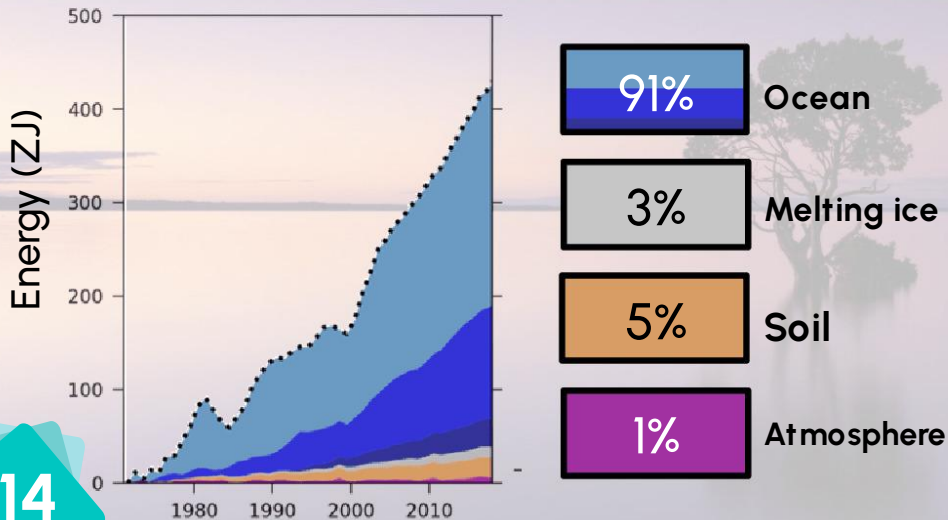
Aerosol Emissions





Despite what it seems, this has nothing to do with spray cans. Aerosols are tiny polluting particles that come from the incomplete combustion of fossil fuels. They are bad for human health and have a negative contribution to radiative forcing (they cool the climate).

Energy Budget

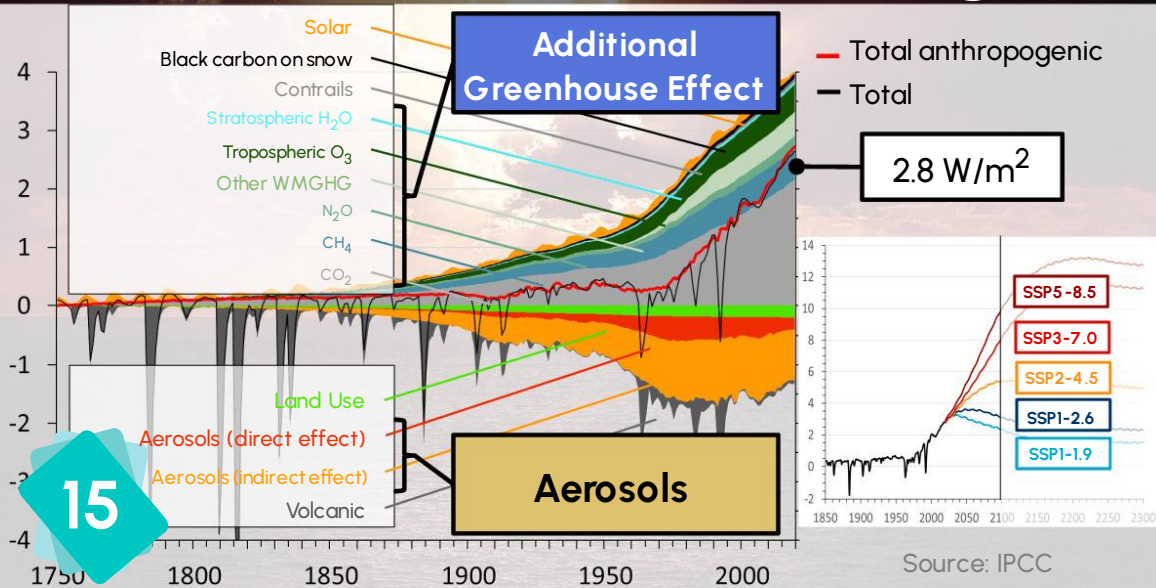


Source: IPCC



Due to radiative forcing, there is extra energy in our planet's system. This graph shows where that energy ends up: It heats the ocean, melts ice, goes into the ground, and warms the atmosphere.

Radiative Forcing





Radiative forcing is the overall balance between the energy coming into the Earth and the energy leaving it. The greenhouse effect is 3.8 W/m^2 (watts per square metre) and the effect of aerosols is -1 W/m^2 , making 2.8 W/m^2 in total.

Melting Glaciers

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Almost every glacier has receded,
and hundreds have already
disappeared. Glaciers are important
because they regulate the supply of
freshwater.

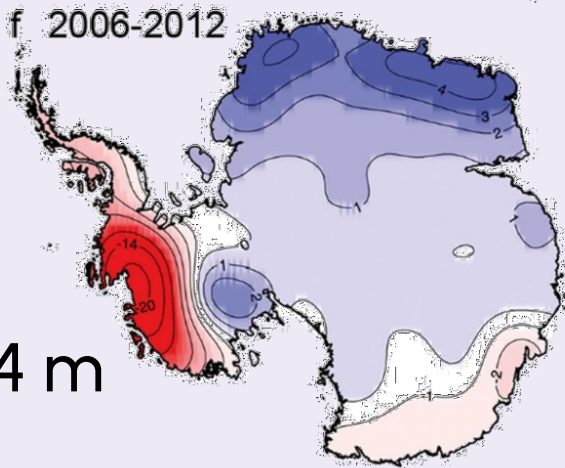
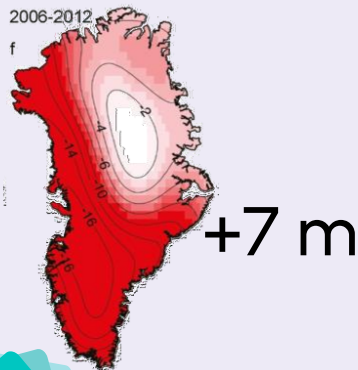
Rising Water Temperature





The ocean absorbs 91% of the energy accumulated on Earth. Like many materials, water expands when it is heated.

Melting Ice Sheets



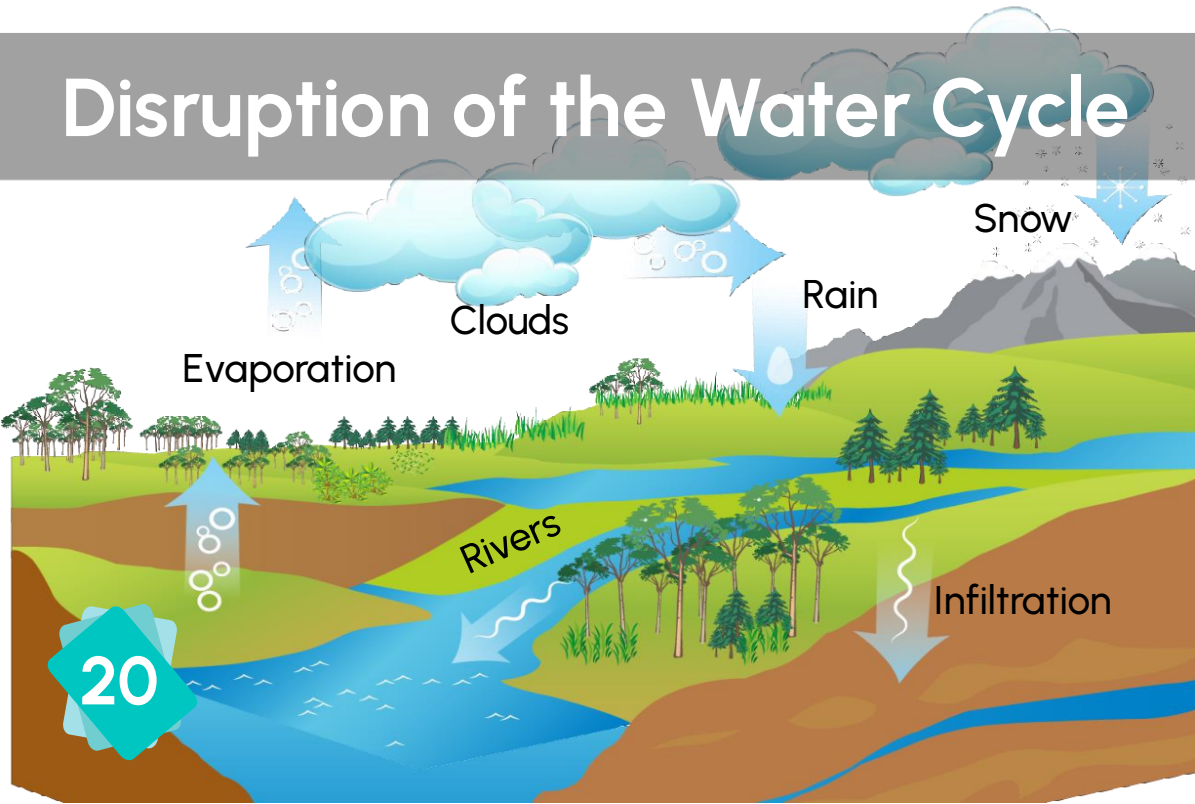
19

Source: IPCC



Greenland and Antarctica are ice sheets (or continental glaciers). If they melted completely, sea levels would rise by 7m from Greenland and 54m from Antarctica. During the last ice age, there were much bigger ice sheets and the sea level was 120m lower than today.

Disruption of the Water Cycle



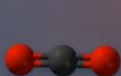
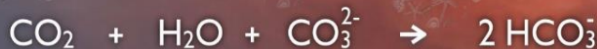


Warmer water and air leads to increased evaporation off the surface of the ocean, making more clouds and rainfall. More evaporation over land dries out the soil.

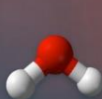
Calcification Difficulties

HOW WILL CHANGES IN OCEAN
CHEMISTRY AFFECT MARINE LIFE?

CO₂ absorbed from the atmosphere



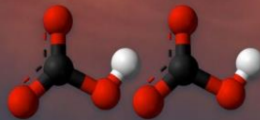
carbon
dioxide



water



carbonate
ion



2 bicarbonate
ions

consumption of carbonate ions impedes calcification



When the pH of the ocean drops, it becomes harder to make calcium carbonate, which is important for making shells.

Terrestrial Biodiversity





The causes of biodiversity loss are: the destruction of habitats, overexploitation of wild species, pollution, climate change, and the introduction of invasive alien species.

When there are changes in the environment, like temperature and water availability, species are forced to migrate and adapt their way of life.

River Flooding

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Disruption of the water cycle can lead to either more or less rainfall.

More rain can cause rivers to flood.

Floods can be worse in urban areas or when soil is very dry, as the water runs off.

Marine Biodiversity

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Pteropods and coccolithophores are at the base of the ocean food chain. If they are driven to extinction, all marine biodiversity will be threatened. Warming waters and overfishing also threaten ocean life.

Cyclones





Cyclones get their energy from warm water at the surface of the ocean. Climate change is making cyclones more powerful.

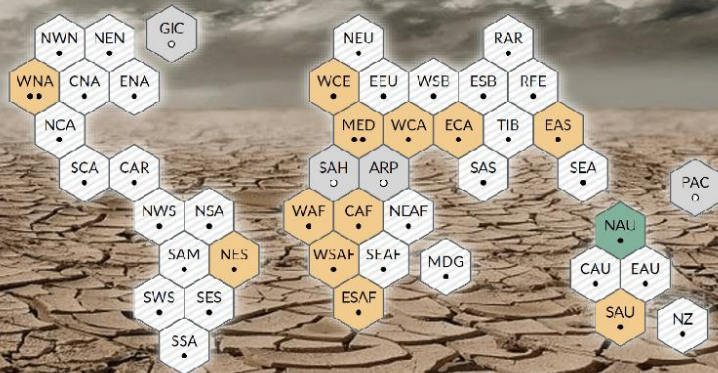
Pteropods and Coccolithophores





Pteropods are a type of zooplankton and coccolithophores are a type of phytoplankton. These organisms have shells made of calcium carbonate.

Droughts





The disruption of the water cycle can bring more or less rainfall. Too little rain can cause a drought. Droughts are likely to become more frequent in the future.

Coastal Flooding



Cyclones and atmospheric disturbances bring wind, waves, and low pressure conditions. Whenever pressure drops by 1 hectopascal, the water level rises by 1cm. These conditions can lead to coastal flooding, which is made worse by rising sea levels.

Carriers of Disease





Global warming causes animals to migrate. Some carry diseases to new areas where people are not immune.

Access to Freshwater





Our access to freshwater is affected by changes in rainfall and by melting glaciers, as glaciers regulate the flow of rivers.

Reduced Crop Yields



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Food production can be affected by temperature, droughts, extreme weather events, and flooding (e.g. the Nile Delta).

Wildfires



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Wildfires start and spread more easily during droughts and heatwaves. They emit CO₂ in the same way as deforestation. This creates a feedback loop.

Heatwaves





Heatwaves are set to become more frequent and more intense. The combination of high temperatures and humidity can create deadly conditions for humans. The number of days when these fatal weather conditions occur is set to rise.

Famine



Reduced crop yields and reduced
marine biodiversity can cause
famines.

Human Health



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Famine, diseases carried by animals, heatwaves, and armed conflicts can affect human health.

Population displacement



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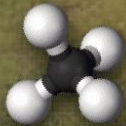
Climate change will inevitably make large areas uninhabitable and people will be forced to move.

Armed Conflicts

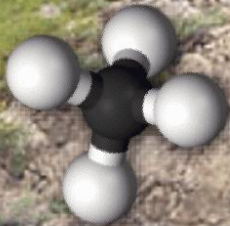
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We shouldn't let it come to this...



Permafrost





Permafrost is permanently frozen ground. As it thaws, it is releasing CO₂ and methane into the atmosphere. These gases come from previously frozen organic matter which starts to decompose. Melting permafrost creates a feedback loop.

Growing Inequality

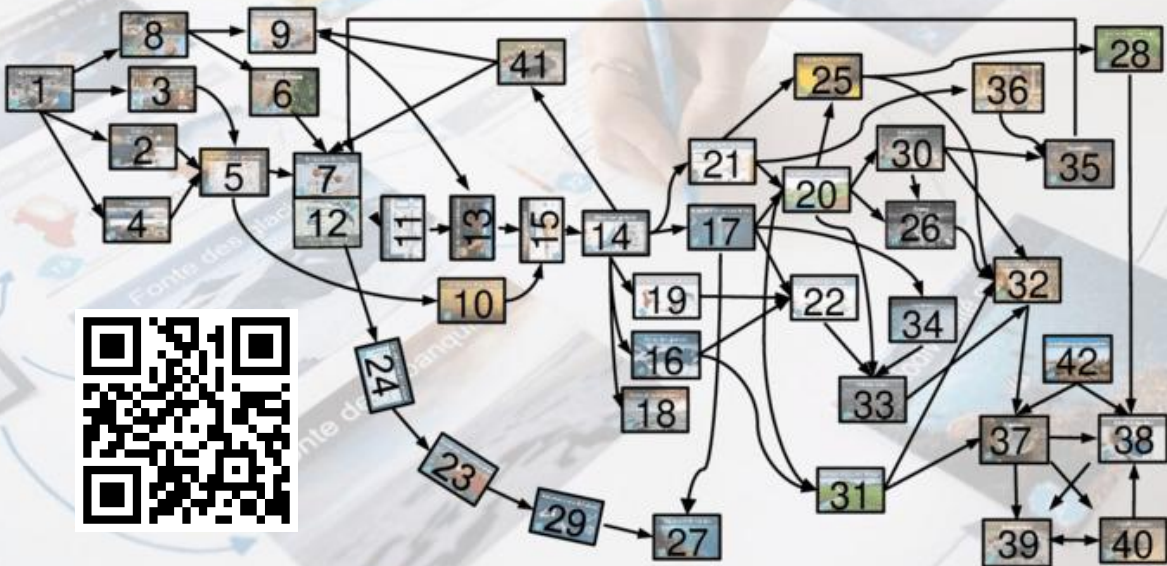


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Climate change reinforces social and economic inequalities: between individuals, between countries, and between generations. To prevent these inequalities from getting worse, we must ensure climate policies are fair and equitable.

Key



The Climate Fresk was developed by Cédric Ringenbach.
Its distribution is managed by the non-profit organisation 'La Fresque du Climat'.

To facilitate a Climate Fresk, you must attend a Climate Fresk workshop and a facilitator training.

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